CASE 19

Improving Road Safety

Vietnam's Comprehensive Helmet Law

The Case at a Glance

HEALTH GOAL: To reduce road-traffic head injuries and deaths.

STRATEGY: Coordinated, multipronged plan building on national Helmet Day to change helmet-wearing behavior at scale, including expanded enforcement and a large financial penalty for noncompliance.

HEALTH IMPACT: 1,557 lives saved and 2,495 serious injuries prevented (2008).

WHY IT WORKED: Careful preplanning and coordination of multiple elements. Strengthened traffic police force for effective enforcement. Hefty fines to deter noncompliance. Widespread compliance with the new law on Helmet Day. High-level government leadership. Hard-hitting information campaigns. Local availability of affordable helmets appropriate to the hot, humid climate.

FINANCING: Cost about US$24 million for new helmets and police salaries. Estimated cost-effectiveness ratio: US$1,248 per disability-adjusted life year averted.

SCALE: National.

Some motorcyclists walk away from a crash with a few scratches and bruises, but others suffer lifelong consequences or even death. Head injuries are common in motorcycle accidents; brain trauma can leave injured motorcyclists with personality changes, cognitive impairment, and disabilities that force them to use a wheelchair for life.

In a motorcycle accident, high-quality helmets can reduce the risk of death by 40 percent and the risk of serious injury by more than 70 percent.[1] Riders who forgo a helmet put their lives at risk with every ride, a common scenario where helmet laws are weak, not comprehensive, or nonexistent. This was the situation in the mid-1990s in Vietnam, where a booming economy prompted many Vietnamese to upgrade from bicycles to motorcycles.[2] As of 2014, more than 40 million motorbikes filled Vietnam's busy streets every day, accounting for nearly 95 percent of all Vietnamese road traffic.[3] And these numbers are rising further as the economy continues to grow.

Mandatory helmet laws can persuade riders to wear helmets and thus reduce the frequency and severity of head injuries.[4] Yet in the early years of the 21st century, despite Vietnam's efforts to require helmet use, less than one-third of motorcyclists were donning helmets.[5] With so many unprotected drivers and passengers on Vietnam's crowded streets, the country had a sky-high traffic fatality rate.[6]

In mid-2007, Prime Minister Nguyen Tan Dung announced a new, simple law: starting December 15 of that year—"Helmet Day"—all motorcycle riders would be required to wear helmets on all roads. Those who did not would be subject to a hefty fine from traffic police. A wide range of partners worked with the national government to put the law in place, communicate its contents, and enforce it, including the World Health Organization (WHO), the Asia Injury Prevention Foundation, and private corporations. Thanks to government leadership, the deterrent effect of fines, and policy changes to close loopholes, more than 90 percent of Vietnamese motorcyclists were wearing helmets by 2013.[7]

This case was originally authored by Miriam Temin, with contributions by Jonathan Passmore.

The Toll of Road Traffic Injuries

The most serious disabilities and deaths following motorcycle crashes are from head and neck injuries.[8] Even after a minor crash, a motorcyclist may feel the effects of a hard knock to the head: severe headache, nausea, dizziness, and blurred vision in the short term; memory loss, depression, and a host of other life-changing problems in the long term.[9] The injured rider may be left unable to perform daily tasks; make decisions; or see, hear, or speak. Traumatic brain damage is often debilitating; it can also be fatal.

The rising toll of motorcycle deaths in low- and middle-income countries is part of a global trend that has accompanied economic growth, increased urbanization, and the desire for personal motorized transport. In some of these countries, head injuries caused nearly 90 percent of motorcycle deaths in 2013.[10] Road traffic injuries account for an increasing proportion of all deaths worldwide.[11] If current trends continue, road traffic deaths are predicted to be the fifth leading cause of death worldwide by 2030.[12] In absolute terms, the total number of road traffic deaths is surprisingly high: around 1.24 million every year—about the same as the number of deaths from tuberculosis and twice the number of deaths caused by malaria.

Vietnam's traffic is among the most dangerous in the world.[13] The roads themselves are extremely hazardous. Dangerous habits such as drunk driving are common, and the roads are packed with drivers and riders who often flout the rules. Important traffic laws are in place but not consistently enforced.[14] In 2007, traffic accidents were the leading cause of death among working-age Vietnamese. That year, around 14,000 people, including 2,000 children, lost their lives in traffic accidents.[15] In addition, traffic accidents caused more than 30,000 head and severe brain injuries. And the real toll may be up to a third higher, given underreporting.[16]

Experts estimated that nearly 60 percent of those who lost their lives were motorcycle riders.[17] Despite their risks, motorcycles had become ubiquitous in Vietnam as the economy took off and the pace of life sped up. From the early 1990s to the early years of the following decade, the number of motorcycles on the road climbed steadily. The recorded number quadrupled in just five years, from 5 million in 2002 to more than 20 million by 2007—about one for every four Vietnamese people.[18] As motorcycle ownership boomed, laws covering helmet use remained out of date.

The toll of motorcycle injuries was emotional as well as economic—especially when accidents ended in death or traumatic brain injury. A study by the Asian Development Bank found that as early as 2003, road accidents and their consequences were costing Vietnam at least US$900 million each year, equal to 2.7 percent of the country's gross domestic product (GDP).[19]

Putting an Idea in Motion: Vietnam Takes Action as Traffic Injuries Take Off

The government had begun to take on motorcycle safety as early as the mid-1990s, yet the toll of motorcycle-related deaths continued to grow. In 2001, the first mandatory helmet use law required helmets on national highways and provincial roads. This covered less than a quarter of the country's road system and excluded cities, where motorcycle use was most common. And the law imposed only a minor fine for noncompliance.[20]

In subsequent years, the government continued to expand helmet legislation, for example, to mandate helmet use on all roads.[21] However, the regulations encountered major implementation and enforcement barriers.[22] Traffic police struggled to keep up with the explosion of motorcycle ownership as drivers routinely disregarded the laws—which was somewhat understandable, given the cost of purchasing a helmet and the discomfort of wearing one in the hot climate. According to one journalist, motorcycle riders dubbed helmets "rice cookers"; even after new legislation took hold, he reported that the government's "warnings, celebrity entreaties, and grainy pictures of people drooling from head injuries just unleashed angry protests by riders."[23] With just a small chance of receiving a minor penalty, most motorcyclists did without.[24] Traffic accidents reached an all-time high in 2002.

Then, in 2002, new leadership in the government's National Traffic Safety Committee (NTSC) and other factors created a window for action. Bui Huynh Long, the incoming director of the NTSC, had come from the Ministry of Transport, where he had a track record of support for helmet legislation.[25] His passionate, committed leadership gave international supporters in the Global Road Safety Partnership and other organizations a solid government partner for collaboration.[26]

Long's arrival at the NTSC coincided with helmet initiatives from several international partners. Among them was the Hanoi-based Asia Injury Prevention (AIP)

Foundation, founded by Greig Craft, an American who suspected that a law would change behavior only if people had helmets they were willing to wear. In 2002 the AIP Foundation had opened a factory in Hanoi that produced high-quality helmets for use in tropical settings. The Vietnam Safety Products and Equipment Company produced roughly half a million Protec Tropical Helmets in its first dozen years of operation.[27]

It took several more years to develop the comprehensive helmet law—"Resolution 32"—but by 2007 it was ready. Prime Minister Dung helped announce the new law, which made helmet use mandatory in Vietnam for all motorcycle drivers and passengers on all roads. The law would go into effect on December 15, 2007, the first-ever Helmet Day.[28] In August, in preparation, the government required all its employees—around four million people—plus members of the armed forces to don helmets.[29]

Unlike previous helmet reforms, this new law had support from the Communist Party. Party leadership had become convinced that improving road safety was a sufficiently urgent goal to require mass action. Provincial party chairs got involved by heading local NTSC chapters, and they became responsible to the government and the party leadership for traffic accidents in their localities.[30] Party leaders and other government workers also signed their own "commitments" to wear helmets themselves and to comply with other traffic safety rules.

Helmet Legislation in Action

Helmet Day signaled the launch of the comprehensive helmet law. The law was unambiguous: all riders and passengers needed to wear helmets on all roads—no exceptions.[31] But the strong legislation needed more than a flashy kickoff to improve compliance. Ongoing enforcement, penalties, adjustments, communications, and measurement were paramount to its successful implementation.

The law gave riders an incentive to comply: a substantial fine more than 10 times previous penalties. If caught, bareheaded motorcyclists were fined VND100,000–VND200,000 (US$6–US$12), one-third of the average monthly income.[32] This could be more than the price of some helmets, making it more economical to buy a helmet than to go without.[33] The traffic police were responsible for catching offenders and received training to do so effectively. In the year after the new law took force, police ticketed nearly 680,000 riders for their failure to comply, generating revenue for the state treasury.[34]

But riders already knew all about the new law. The groundwork for Helmet Day had been laid earlier in 2007 by a public education campaign spearheaded by the AIP Foundation with Ogilvy & Mather (Vietnam) dubbed "Enough Is Enough" and "No Excuses . . . Wear a Helmet." The hard-hitting, award-winning campaign used concerts, billboards, and television commercials to educate people about the benefits of helmets and the penalties for noncompliance.[35] The public education campaign didn't focus on helmet safety basics—riders already understood the dangers. Instead it focused on the "real" reasons for noncompliance: "It ruins my hair" and "It will look awful with my trendy clothes." The campaign countered these complaints by demonstrating that the effects of a serious head injury look far worse.[36]

The motorcycle helmet market expanded quickly as demand grew, but quality did not follow suit. Young people were soon seen wearing trendy yet flimsy helmets with cartoon and sports logos on them, "and equally dubious products with special holes at the back for . . . ponytails."[37] Helmet quality is still a major concern. A 2012 study found that fewer than one in five helmets passed safety tests, despite stricter regulatory standards starting in 2008.[38]

With the jump in helmet use, riders found new loopholes to exploit. For example, many riders would wear helmets but leave the chin straps unhooked. The government caught on, and in 2008 a new policy imposed a penalty for an unhooked strap, equal to the fine for riding without a helmet.[39]

Another challenge was presented by earlier legislation, which made it illegal to fine children under 16 for not wearing a helmet.[40] (About 80 percent of urban Vietnamese children rode motorcycles as passengers.[41]) Again, the government adjusted its policy: in 2010 it introduced helmet legislation making helmets mandatory for children ages six and older, with a fine for noncompliance.[42] Acquiring the helmets created a financial burden for some families, and the AIP Foundation stepped in to help families comply, distributing more than a quarter million free child-sized helmets through its Helmets for Kids campaign.[43]

The Payoff: Safer Riders, Fewer Head Injuries

Observational studies confirm that widespread behavior change followed the launch of Resolution 32. In the first days following Helmet Day, police and volunteers from the Youth Union monitored compliance on street corners;

95,000 commune police workers stepped in to help.[44] Starting in 2007, researchers from the Hanoi School of Public Health and the WHO collaborated to track helmet use before and after the law. They observed more than half a million motorcycle riders and passengers in 45 randomly selected sites across three provinces. Before Helmet Day in 2007, 40 percent of the observed riders wore helmets. By early 2011, they found that 93 percent of all observed riders were wearing helmets.[45]

Although Resolution 32 was released without specific plans for an impact evaluation, the Ministry of Health established a hospital-based surveillance system to track road traffic injuries and deaths. The data suggest that the helmet law and related initiatives achieved public health success. Researchers from the WHO and Ministry of Health estimate that Vietnam saw a 16 percent reduction in road traffic head injuries and an 18 percent reduction in road traffic deaths in the three months after the law's enactment.[46] Unfortunately, several hospitals stopped participating in the surveillance system over time, limiting analysis of the law's long-term impact.[47] According to ongoing surveillance at Hanoi's Viet Duc Hospital, one of the country's largest surgical centers, there were nearly 8 percent fewer head injuries among traffic-accident patients in 2010 than before the law, a statistically significant difference.[48]

The police also monitored implementation of the law. National police data suggest that Resolution 32 saved 1,600 lives and averted 2,500 serious injuries in its first year.[49] Given that police data often miss unreported injuries and fatalities, the real impact may be greater.[50] Another assessment, undertaken by the AIP Foundation, suggests that the law prevented 20,609 deaths and 412,175 serious injuries from 2008 to 2013 (see Box 1).[51]

Since these data were not generated through experimental methods, they should be interpreted with caution. Nonetheless, the evidence of declining head injuries in the context of ever-expanding motorcycle ownership strongly suggests that Resolution 32 prevented more injuries and saved more lives than Vietnam's previous helmet legislation.

Resolution 32 also helped Vietnam outperform its neighbor, Cambodia. Cambodia's 2009 law made helmets mandatory for motorcycle drivers—but not for their passengers. As of 2010, only 65 percent of Cambodian drivers and 9 percent of passengers wore helmets, and Cambodia had the highest vehicular fatality rate of any country in Southeast Asia.[52] In 2014 the Cambodian government expanded its helmet law to cover passengers.

Box 1. Strength of the Evidence

It is common sense that helmets protect motorcyclists' heads during a crash, so it is no surprise that evidence from a handful of countries suggests that laws to mandate helmet use decrease head injuries and death. However, high-quality studies of national road traffic measures are hard to find, and methodological challenges limit the scope for quasi-experimental designs. A Cochrane systematic review found 61 studies, but none were randomized, most relied on observational designs, and all but a handful focused on helmet laws in high-income countries.[53]

The question remains: can national-level behavior change sustain the consistent, correct use of motorcycle helmets? In Vietnam, the main source of potential data is a hospital-based surveillance system of road traffic injuries, put in place by the Ministry of Health in 2007. Using a before-and-after study design in which numbers of injuries were assessed three months before and after Helmet Day, the system drew data from a total of 20 provincial and central hospitals around the country.[54] The sample consisted of more than 80,000 injury patients (injuries that resulted in death were counted only if death occurred after the accident, in the hospital, not at the scene of the accident). The data suggest that the risk of head injury and death as a result of a traffic accident decreased following the introduction of the law, by 16 percent and 18 percent, respectively.[55] Another study, by the AIP Foundation, estimated a much larger impact.[56]

Several limitations prevent full confidence in the results of the analysis. Hospitals counted all road traffic injuries, not just those among motorcycle riders, and did not record whether the patients had been wearing helmets or whether there were any weather-related variations. There was no validation of staff reporting to prevent undercounting, biased reporting, or other inconsistencies in data collection. Other estimates also face limitations because of the risks of underreporting.

Gains at What Price?

Brain injury is the most common injury from motorcycle crashes, and hospital care to treat such injuries can be expensive.[57] One study found that the average medical costs in Vietnam for a person suffering from a severe head trauma came to US$2,370, which exceeded the national GDP per capita of US$1,910 in 2013.[58] Treatment for catastrophic conditions can impoverish entire families and thus magnify the economic effects of a crash.[59] As many as 70 percent of traffic accident victims lose income, and two-thirds of them must take on debt to compensate.[60]

Studies from countries of all income levels have found significant evidence that mandatory helmet laws, combined with enforcement, can greatly reduce the burden of traffic injuries on families, the health system, and the economy. They conclude that legislation is among the most cost-effective interventions available.[61]

In the first year after passage of the decree, researchers for *Millions Saved* estimate that the law averted 90,582 disability-adjusted life years (DALYs) from prevented deaths and nonfatal injuries.[62] This is a conservative estimate because prevented deaths are likely to be concentrated among young adult males; their deaths or permanent injuries would result in more life years lost than death or injury incurred by the average, older person on the standard age distribution. In addition, number of deaths is estimated from police data—a source that is subject to underreporting. The estimated cost of the initiative was the cost of helmets and of hiring new police officers to enforce the law. The calculation yielded a cost-effectiveness ratio of US$1,248 per DALY averted.

This ratio falls above the range reported elsewhere, of US$467 to US$769 per DALY averted, likely because the data source and methods used underestimated impact. Nonetheless, the cost-effectiveness ratio is close to the GDP per capita for Vietnam.

The Keys to Lasting Success

The success of the 2007 law was made possible by strong leadership, a vibrant partnership, a hefty fine for noncompliance, and culturally appropriate approaches.[63] The prime minister and his party threw their weight behind simple legislation that reduced confusion about enforcement: all riders and passengers were covered on all roads. Since then, policy adjustments have closed loopholes and given the helmet law sharper teeth.

Resolution 32 had broad-based support across the Communist Party, the government, and a range of other organizations, and communication campaigns helped it take hold across the country. Political leaders featured prominently in media spots to promote the new legislation, supported by groups like the Youth Union.[64] Traditional and new international partners supported the government's effort. The AIP Foundation's engagement with the Global Road Safety Partnership, initiated by the World Bank, brought private-sector funding from automobile manufacturers Ford Motors and Toyota.[65] Meanwhile, Bloomberg Philanthropies supported social marketing and legislative action to increase police enforcement.[66]

The hefty fine was one of the law's essential design elements. Even with the considerable effort put into other activities, the fear of getting a penalty may have been riders' deciding factor. The consistency of enforcement by traffic police made the threat feel real and helped spur compliance.[67] The paperwork was a nuisance, and paying the fine required taking time off from work. A further disincentive was that riders stopped by the police for not wearing a helmet also risked being cited for other traffic violations.

To achieve behavior change, planners knew that their approach had to be culturally appropriate. In Vietnam, where conformity is highly valued, planners leveraged the power of conformity to ensure adoption of the helmet decree. Through the population-wide launch, everyone adopted the new behavior all at once, thus motivating helmet wearing through peer pressure.[68] It worked: helmet use seems to have become a new social norm.

To be able to comply with the law, the Vietnamese also needed access to affordable helmets. High-quality, climate-appropriate helmets were available to those who could afford them, and the government had progressively strengthened helmet quality standards as of 2008. But low-quality helmets, some costing as little as US$2.40, still flooded the market.[69] Identifying and discouraging the sale and use of substandard helmets remained a major challenge.[70] The NTSC and its partners have worked hard to take substandard helmets off the streets, and their efforts are beginning to pay off. Trade and customs officials are clamping down on the import of Chinese helmets, whose manufacturers falsely claim that their products adhere to quality standards.[71]

Children's safety is still a serious concern. Initial legislative inconsistencies made it easy for helmeted parents to tote around unprotected kids. Even when the policy changed to cover children over the age of six, many parents still failed to put helmets on their children's heads. As of 2010, just 18 percent of primary school children wore helmets while riding as motorcycle passengers in major cities.[72] Partly to blame is a widely quoted media report that erroneously suggested that helmets injure children's necks.[73] Despite the lack of evidence, this common myth deters well-intentioned parents from protecting their small children. The Helmets for Kids campaign, led by the AIP Foundation, attempts to counteract this false claim through education and donations of children's helmets.

Implications for Global Health

Vietnam's helmet law experience suggests that a law can successfully change behavior—perhaps especially in a country where the government exercises strong control, as in Vietnam. But even there, the law needed effective enforcement with high penalties to ensure compliance.

The government still faces several persistent challenges: children's helmet wearing, helmet quality, and correct usage. There are clear steps the government can take to overcome these challenges. People who cannot afford helmets need donated ones, and babies and young children need small tropical helmets. And everyone needs a helmet that meets the national safety standard. Substandard helmets must attract the same penalty as no helmet, and riders must learn that chin straps are essential to their protection in a crash.[74]

The AIP Foundation is helping other countries learn about the lessons from Vietnam. It now has offices around the region, in Cambodia, China, and Thailand, and has expanded to Uganda and Tanzania. Its factory has sparked particular interest. A delegation of government officials from Cambodia visited the facility and considered plans to open an in-country helmet plant to support their country's new passenger helmet requirement.[75]

Legislation mandating helmet use has advanced globally, yet only one-third of countries rate enforcement of their helmet laws as "good."[76] Encouraging helmet wearing is a central goal of the Global Plan for the Decade of Action for Road Safety,[77] and a 2014 United Nations General Assembly resolution, "Improving Global Road Safety," asks member states to enact legislation on motorcycle helmets and take other measures to reduce risk.[78] One clear priority: to collect and disseminate more and better evidence from low- and middle-income countries on how to improve road traffic safety at scale.

REFERENCES

ADB (Asian Development Bank). 2012. *Road Safety Action Plan.* Mandaluyong, the Philippines: ADB. http://www.adb.org/sites/default/files/institutional-document/33427/files/road-safety-action-plan.pdf.

ADB and ASEAN (Association of Southeast Asian Nations). 2003. *Regional Road Safety Program Accident Costing Report: The Cost of Road Traffic Accidents in Vietnam.* Manila, the Philippines: ABD and ASEAN.

AIP (Asia Injury Prevention) Foundation. 2013. "Trade Your Used Helmet for a Standard Quality Helmet at Protec." News release, March. http://aip-foundation.org/wp-content/uploads/2013/05/Protec-Exchange-Program-March-2013-English.pdf.

———. 2014. *Developing an Integrated Campaign to Address Child Helmet Use in Vietnam: A Case Study.* New York: Atlantic Philanthropies. http://issuu.com/aipfoundation/docs/case_study_-_vietnam_national_helme.

Asian Scientist. 2012. "Vietnam's Baseball Cap 'Helmets,'" December 3. http://www.asianscientist.com/2012/12/health/vietnam-baseball-cap-helmets-2012/.

Bishai, David M., and Adnan A. Hyder. 2006. "Modeling the Cost Effectiveness of Injury Interventions in Lower and Middle Income Countries: Opportunities and Challenges." *Cost Effectiveness and Resource Allocation* 4 (2).

Bloomberg Philanthropies. *2013 Leading the Worldwide Movement to Improve Road Safety.* New York: Bloomberg Philanthropies. http://www.mikebloomberg.com/content/uploads/sites/10/2015/06/Bloomberg_Philanthropies_Leading_the_Worldwide_Movement_to_Improve_Road_Safety.pdf.

Britton, Eric. 2009. "Lessons from a Helmet Wearing Campaign in Vietnam." *World City Bike Forum,* February 6. https://groups.yahoo.com/neo/groups/WorldCityBike/conversations/topics/597.

Center for Injury Prevention and Policy Research. 2008. *Study Report on Traffic Injury Situation in Provinces of Yen Bai, Da Nang and Binh Duong.* Hanoi, Vietnam: Ha Noi School of Public Health.

Dribben, Melissa. 2008. "Vietnam Shows Effect of Motorcycle Helmets: Injuries Dropped up to 30% after Their Use

REFERENCES, *continued*

Became Mandatory." Philly.com, June 28. http://articles.philly.com/2008-06-28/news/25249036_1_helmet-law-head-injury-deaths-helmet-legislation.

Hansen, Arve. 2014. "Hanoi's Looming Traffic Nightmare." *Diplomat,* September 7. http://thediplomat.com/2014/09/hanois-looming-traffic-nightmare/.

Hiep, Dinh Van, and Pham Thi Quyen. 2009. "Road Traffic Accidents in Vietnam." *Road Research* (January): 6. http://www.academia.edu/209613/Analysis_of_road_traffic_accidents_in_Vietnam.

Hill, P.S., A.D. Ngo, T.A. Khuong, H.L. Dao, H.T.M. Hoang, H.T. Trinh, L.T.N. Nguyen, and P.H. Nguyen. 2009. "Mandatory Helmet Legislation and the Print Media in Viet Nam." *Accident Analysis and Prevention* 41 (4): 789–97.

Hoang, Hanh T.M., Tran L. Pham, Thuy T.N. Vo, Phuong K. Nguyen, Christopher M. Doran, and Peter S. Hill. 2008. "The Costs of Traumatic Brain Injury Due to Motorcycle Accidents in Hanoi, Vietnam." *Cost Effectiveness and Resource Allocation: C/E* 6 (August): 17. doi:10.1186/1478-7547-6-17.

Huu, Nguyen, Duong Thi Mai, Nguyen Thien, and Nguyen Ngoc. 2011. "Study on Vietnam Traffic Accident Situation via Data of Health Sector." *Journal of the Eastern Asia Society for Transportation Studies* 9: 2011–21.

Hyder, A.A., H. Waters, T. Phillips, and J. Rehwinkel. 2007. "Exploring the Economics of Motorcycle Helmet Laws Implications for Low and Middle-Income Countries." *Asia-Pacific Journal of Public Health* 19 (2): 16–22. doi:10.1177/10105395070190020401.

Le, Duy. 2008. "Mother Weighs Dangerous Options." Vietnam News, January 30. http://vietnamnews.vn/talk-around-town/173392/mother-weighs-dangerous-options.html.

Le, L., C. Pham, M. Linnan, V.D. Dung, N.L. Phuong, H.L. Hanh, and V.L. Anh. 2002. "Vietnam Profile on Traffic-Related Injury: Facts and Figures from Recent Studies and Their Implications for Road Traffic Injury Policy." Paper presented at the Road Traffic Injuries and Health Equity Conference, Cambridge, MA, April 10–12.

Liu, Bette C., Rebecca Ivers, Robyn Norton, Soufiane Boufous, Stephanie Blows, and Sing Kai Lo. 2008. "Helmets for Preventing Injury in Motorcycle Riders." *Cochrane Database of Systematic Reviews* 2008 (1): Article CD004333. http://doi.wiley.com/10.1002/14651858.CD004333.pub3.

Magnier, Mark. 2010. "Helmets Are a Fashion Must-Have in Vietnam; The Law Says So." *Los Angeles Times,* May 25. http://articles.latimes.com/2010/may/25/world/la-fg-vietnam-helmets-20100526.

Mason, Margie. 2007. "Vietnam Enacts Motorbike Helmet Law." *USA Today,* October 3. http://usatoday30.usatoday.com/news/world/2007-10-03-469093791_x.htm.

Mayo Clinic Staff. 2015. "Diseases and Conditions: Traumatic Brain Injury." Mayo Clinic, May 15. http://www.mayoclinic.org/diseases-conditions/traumatic-brain-injury/basics/symptoms/con-20029302.

McDonnell, Mary Byrne, Van Bich Thi Tran, and Nina R. McCoy. 2010. *Helmet Day: Lessons Learned on Vietnam's Road to Healthy Behavior.* Brooklyn, NY: Social Science Research Council.

Minh, Hung. 2014. "Vietnam Shares Road Traffic Safety Experiences with Cambodia." *Thanhnien Daily,* October 17. http://www.thanhniennews.com/society/vietnam-shares-road-traffic-safety-experiences-with-cambodia-32688.html.

Mirelman, Andrew, Amanda Glassman, and Miriam Temin. 2016. *Estimating the Avertable Disease Burden and Cost-Effectiveness in* Millions Saved *Third Edition.* CGD Working Paper. Washington, DC: Center for Global Development.

National People's Assembly of Socialist Republic of Viet Nam. 2002. *Ordinance on Handling of Administrative Violations.* Ordinance No. 44/2002/PL-UBTVQH10, July 2. http://www.wipo.int/wipolex/en/details.jsp?id=6682.

Ngo, Anh D., Chalapati Rao, Nguyen Phuong Hoa, Damian G. Hoy, Khieu Thi Quynh Trang, and Peter S. Hill. 2012. "Road Traffic Related Mortality in Vietnam: Evidence for Policy from a National Sample Mortality Surveillance System." *BMC Public Health* 12 (1): 561. doi:10.1186/1471-2458-12-561.

Nguyen, Ha Trong, Jonathon Passmore, Pham Viet Cuong, and Nam Phuong Nguyen. 2013. "Measuring Compliance with Viet Nam's Mandatory Motorcycle Helmet Legislation." *International Journal of Injury Control and Safety Promotion* 20 (2): 192–96. doi:10.1080/17457300.2012.706617.

O'Flaherty, Bridget. 2012. "Traffic: Vietnam's Silent Killer." Diplomat, August 28. http://thediplomat.com/2012/08/traffic-vietnams-silent-killer/.

Olson, Zachary, John A. Staples, Charles Mock, Nam Phuong Nguyen, Abdulgafoor M. Bachani, Rachel Nugent, and Stéphane Verguet. 2016. "Helmet Regulation in Vietnam: Impact on Health, Equity and Medical Impoverishment." *Injury Prevention.* doi:10.1136/injuryprev-2015-041650.

Passmore, Jonathon W., Lan Huong Nguyen, Nam Phuong Nguyen, and Jean-Marc Olivé. 2010. "The Formulation and Implementation of a National Helmet Law: A Case Study from Viet Nam." *Bulletin of the World Health Organization* 88 (10): 783–87. doi:10.2471/BLT.09.071662.

Pham, K.H., Q.X. Le Thi, D.J. Petrie, J. Adams, and C.M. Doran. 2008. "Households' Willingness to Pay for a Motorcycle Helmet in Hanoi, Vietnam." *Applied Health Economics and Health Policy* 6 (2–3): 137–44.

Prime Minister. 2012. *Approval of the National Road Safety Strategy by 2020 and a Vision to 2030.* Decision 1586/QD-TTg. http://www.who.int/roadsafety/decade_of_action/plan/vietnam_plan.pdf?ua=1.

REFERENCES, *continued*

Protec. 2011. "Helmet Structure: Frequently Asked Questions," October 17. http://protec.com.vn/web/en/frame/faq_category/q-a.html.

Stokes, Connla. 2012. "Letter from Vietnam: Hard-Headed Motorcyclists." *Guardian*, March 20. http://www.theguardian.com/world/2012/mar/20/letter-from-vietnam-motorcycle-helmets.

Talk Vietnam. 2010. "Elevating Awareness of Need for Helmets among Children," September 16. http://www.talkvietnam.com/2010/09/elevating-awareness-of-need-for-helmets-among-children/.

United Nations General Assembly. 2014. *Improving Global Road Safety.* New York: United Nations General Assembly. http://www.who.int/roadsafety/news/2014/Final_draft_UN_General_Assembly_resolution_improving_global_road_safety.pdf?ua=1.

US CDC (Centers for Disease Control and Prevention). n.d. *Motorcycle Safety: How to Save Lives and Save Money.* Atlanta, GA: CDC. http://www.cdc.gov/motorvehiclesafety/pdf/mc2012/motorcyclesafetybook.pdf.

WHO (World Health Organization). 2006. "How to Design and Implement a Helmet Programme." In *Helmets: A Road Safety Manual for Decision-Makers and Practitioners.* Geneva: WHO. http://www.who.int/roadsafety/projects/manuals/helmet_manual/3-How%20to.pdf?ua=1.

———. 2013. *Global Status Report on Road Safety 2013: Supporting a Decade of Action.* Geneva: WHO. http://www.who.int/violence_injury_prevention/road_safety_status/2013/en/.

———. n.d. *Global Plan for the Decade of Action for Road Safety 2011–2020.* Geneva: WHO. http://www.who.int/roadsafety/decade_of_action/plan/en/.

World Bank. 2015. Data. Accessed November 10. http://data.worldbank.org/.

———. 2016. "GDP per Capita (Current US$)." Data. Accessed February 8. http://data.worldbank.org/indicator/NY.GDP.PCAP.CD.

ENDNOTES

1. Liu et al. (2008).
2. McDonnell, Thi Tran, and McCoy (2010); Hoang et al. (2008).
3. Hansen (2014).
4. US CDC (n.d.); Liu et al. (2008).
5. Passmore et al. (2010).
6. WHO (2013).
7. Nguyen et al. (2013).
8. WHO (2013).
9. Mayo Clinic Staff (2015).
10. WHO (2013).
11. Hoang (2015).
12. Hoang (2015).
13. Dribben (2008).
14. O'Flaherty (2012).
15. Center for Injury Prevention and Policy Research (2008).
16. Le et al. (2002).
17. Le et al. (2002).
18. McDonnell, Thi Tran, and McCoy (2010).
19. ADB and ASEAN (2003), quoted in McDonnell, Thi Tran, and McCoy (2010).
20. McDonnell, Thi Tran, and McCoy (2010).
21. WHO (2006).
22. Hill et al. (2009).
23. Magnier (2010).
24. McDonnell, Thi Tran, and McCoy (2010).
25. McDonnell, Thi Tran, and McCoy (2010).
26. Hill et al. (2009).
27. Hoang (2015). See the Protec website for technical and other information on the company's helmets (Protec 2011); AIP Foundation (2013).
28. McDonnell, Thi Tran, and McCoy (2010).
29. McDonnell, Thi Tran, and McCoy (2010).
30. McDonnell, Thi Tran, and McCoy (2010).
31. Passmore et al. (2010).
32. Passmore et al. (2010).
33. Pham et al. (2008).
34. Passmore et al. (2010).
35. AIP Foundation (2013); Britton (2009).
36. Mason (2007).
37. Stokes (2012).
38. Passmore et al. (2010); *Asian Scientist* (2012).
39. Passmore et al. (2010).
40. National People's Assembly of Socialist Republic of Viet Nam (2002).
41. Talk Vietnam (2010).
42. McDonnell, Thi Tran, and McCoy (2010).
43. McDonnell, Thi Tran, and McCoy (2010).
44. McDonnell, Thi Tran, and McCoy (2010).
45. Nguyen et al. (2013).
46. Based on risk ratios (Passmore et al. 2010).
47. Jonathan Passmore, personal communication with the author, April 10, 2015.
48. Jonathan Passmore, personal communication with the7author, April 10, 2015.
49. Passmore et al. (2010).
50. Huu et al. (2011).
51. AIP Foundation (2014).
52. WHO (2013); Hiep and Thi Quyen (2009).
53. Liu et al. (2008).
54. Passmore et al. (2010).
55. Passmore et al. (2010).
56. AIP Foundation (2014).
57. Hoang et al. (2008).
58. World Bank (2015, 2016).
59. Hill et al. (2009).

ENDNOTES, *continued*

60. ADB (2012).
61. Hyder et al. (2007).
62. Estimation of total DALYs averted was calculated for injuries and deaths from traffic accidents. The helmet policy was found to avert 1,557 deaths and 2,495 injuries (Passmore et al. 2010). The costs of implementing the policy were calculated as US$24 million, including police salaries and costs of helmets (Bishai and Hyder 2006; Olson et al. 2016). The cost-effectiveness is US$1,248 per DALY averted. This puts it right around the GDP per capita of Vietnam in 2008. Adding in treatment costs averted means that there are likely to be many cost savings. See Mirelman, Glassman, and Temin (2016).
63. Hill et al. (2009).
64. Hill et al. (2009).
65. McDonnell, Thi Tran, and McCoy (2010).
66. Bloomberg Philanthropies (2013).
67. AIP Foundation (2014).
68. McDonnell, Thi Tran, and McCoy (2010).
69. Passmore et al. (2010).
70. Stokes (2012); *Asian Scientist* (2012).
71. Jonathan Passmore, personal communication with the author, April 10, 2015.
72. AIP Foundation (2014).
73. For example, see Le (2008).
74. WHO (2013).
75. Minh (2014).
76. WHO (2013).
77. WHO (n.d.).
78. United Nations General Assembly (2014).